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Balancing co-research in policy and practice – insights from a Swedish research council

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Abstract
This study aims to advance understanding of the relationship between policies and practices in public funding of co-research, in view of the ongoing academic reorientation towards a wider and deeper societal engagement – seen as an expression of public sector innovation. Based on a review of three calls by the Swedish Research Council for Health, Working Life and Welfare (Forte), similarities and differences regarding concepts, motives, stakeholders and methods for co-research are distinguished in call texts and applications. The results reveal an ambiguous relationship between co-research policies and practices, with a high compliance among applicants with the calls' request for co-research but with little guidance on how to esteem, prioritize, design and conduct co-research. This complicates the balancing act between the co-existing rationales of academic freedom and public interest in public research funding as part of the two-directional relationship between innovative institutions, processes and solutions in public sector innovation.

Keywords
co-production, co-research, impact, innovation, research funding

Sammanfattning
Denna studie syftar till att öka förståelsen av relationen mellan politik och praktik i offentlig finansiering av samskapande forskning, i ljuset av den pågående akademiska omorienteringen mot ett bredare och djupare samhällsengagemang – som ett uttryck för innovation i offentlig sektor. Baserat på en granskning av tre utlysningar från Forskningsrådet för hälsa, arbetsliv och välfärd (Forte) identifieras likheter och skillnader avseende begrepp, motiv, samhällsaktörer och metoder för samskapande forskning i utlysningstexter och ansökningar. Resultaten visar på ett tvetydigt förhållande mellan policy och praktik, med en hög efterlevnad bland sökande av utlysningarnas efterfrågan på samskapande forskning, men med lite vägledning om hur sådan forskning kan värderas, prioriteras, utformas och genomföras. Detta försvårar balansgången mellan de samexisterande

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rationaliteterna av akademisk frihet och allmänintresse i offentlig forskningsfinansiering, som en del av den dubbelriktade relationen mellan innovativa institutioner, processer och lösningar i offentlig innovation.

Nyckelord
forskningsfinansiering, samhällsnytta, samskapa, samverka

Introduction

Many higher education institutions (HEIs) in the Nordic countries – as well as globally - are currently reorienting themselves towards a wider and deeper societal engagement (Brundenius et al., 2017). This has been discussed in terms of "the developmental university" being "committed in diverse ways to social inclusion through knowledge and, more generally, to the democratization of knowledge" (Arocena & Sutz, 2017, p. 60), as well as "the engaged university" with academic engagement "on a wide range of topics with local communities, using these communities as labs to test new ideas and find better ways to achieve social and economic goals" (Breznitz & Feldman, 2012, p. 141). As part of this reorientation, there is a growing interest among researchers and public research funders in Sweden and other Nordic countries of improving the relevance and impact of research in society, by addressing societal challenges through development of novel knowledge and measures (Brundenius et al., 2017; Lindberg, 2018). Often, this is done together with stakeholders from the public sector, civil society or industry, by means of participatory research approaches – here called co-research (cf. Aagaard Nielsen & Svensson, 2006; Gunnarsson et al., 2016).

Taken together, this reorientation among HEIs, researchers and research funders can be seen as an expression of public sector innovation (cf. Fuglsang et al., 2018; Rønning et al., 2022), in the sense of developing and testing novel policies and practices for societal engagement in publicly funded research. In light of previous research published in the *Nordic Journal of Innovation in the Public Sector*, the reorientation reflects three interrelated levels of public sector innovation: 1) Innovation of public policy, regulation and services, 2) New and innovative processes of collaborative interaction through which innovative solutions are generated, and 3) New and innovative institutional designs that facilitate the scaffolding of these processes (Sørensen & Torfing, 2022). This relies, in turn, on theories of governance, suggesting a two-directional relationship between innovative institutions, processes and solutions in public sector innovation (Ansell & Torfing, 2022; Sørensen & Torfing, 2022).

In this study, we take a closer look at an empirical example of reorientation towards a wider and deeper societal engagement in academic policies and practices of co-research. The example is a learning process about co-research conducted between 2023 and 2026 by the Swedish Research Council for Health, Working Life and Welfare (Forte) in collaboration with the Forum for Social Innovation Sweden (FSIS) at Malmö University, with which two of the present authors are affiliated. Forte encourages stakeholder collaboration (*samverkan*) in all of their calls for research funding, by asking for descriptions of it in application forms and including it in evaluation criteria. Sometimes, Forte encourages a deeper form of collaboration, called co-production (*samskapande*), where stakeholders are actively involved in several parts of the research process. Stakeholders here refer both to individuals, in their capacity as citizens, users, professionals, and so on, and to organizations from the public sector, civil society or industry. The stated rationale behind Forte's encouragement of co-research is the belief

that it helps improve the societal relevance and impact, as well as the scientific quality, of their funded research.

Forte's learning process about co-research focuses on three of their calls for funding where stakeholder collaboration and co-production was especially encouraged: Applied welfare research, 2022, Children's and young people's mental health, 2022, and Co-production for a sustainable working life, 2023. The learning process was initiated with a review of co-research in the call texts and applications received. In this article, results from the review are presented and discussed, with the aim of advancing understanding of the relationship between policies and practices in public promotion of co-research. Policies here refer to the formulation and communication of calls for research funding, while practices refer to the planning and design of research studies in applications for funding from these calls. The focus on the relationship between policies and practices is motivated by the multi-level character of public sector innovation, described above (cf. Sørensen & Torfing, 2022). This relationship is studied by identifying similarities and differences between demands for co-research in Forte's three calls for funding and the planned co-research in applications to these calls, with regard to concepts, motives, stakeholders and methods for co-research. Given that co-research is requested to different extents and in different ways across the three calls, they are well suited for a comparative examination. The main method is a document analysis of call texts and applications, analyzed in light of previous research on different levels of co-research policy and practice.

This topic holds scientific and societal relevance, as it addresses the balancing act between promoting scientific excellence and ensuring societal usefulness in public research funding. In Sweden, HEIs are legally bound to "operate under the general principle that academic freedom must be promoted and protected" as well as to pursue "collaboration for mutual exchanges with the surrounding community" (The Swedish Higher Education Act 1992:1434, paragraphs 1.6 and 1.2). As a member of the European Union, Sweden is part of its dual policy advocacy of excellent and open science, where the latter includes public engagement in research to improve its societal relevance and impact (European Commission, 2016). By examining the relationship between policies and practices in public promotion of co-research, this article adds to previous research on both the balancing act between academic freedom and public interest in public research funding (cf. Bégin-Caouette et al., 2017; Lepori et al., 2023) and multi-level relationships in public sector innovation (cf. Sørensen & Torfing, 2022).

Theoretical framework

The study is guided by previous research on co-research policy and practice, that is combined to help capture the multi-level character of public sector innovation, in line with governance theory (cf. Ansell & Torfing, 2022; Sørensen & Torfing, 2022). More specifically, we combine research on research policy rationales (de Jong et al., 2016; Lepori et al., 2023), different modes of knowledge production (Carayannis & Campbell, 2006; Nowotny et al., 2001) and degrees of stakeholder participation (Science Europe, 2017).

To situate co-research in a broader scientific landscape, we first turn to the different modes of academic knowledge production distinguished in research (Carayannis & Campbell, 2006; Nowotny et al., 2001). The three main modes are:

- *Mode 1* - where research is conducted independently by researchers, based on research questions that are relevant primarily from a scientific perspective.

- *Mode 2* - where research is conducted in collaboration with stakeholders, based on research questions that are both theoretically and practically relevant.
- *Mode 3* - where research is conducted in complex systems of different societal actors, based on research questions that help understand and address societal challenges.

Even if these modes are ideal types that partly overlap in practice, they help distinguish co-existing rationales in research and research funding. Since the Second World War, many governments have incorporated research policy and funding as a core task in pursuit of multiple policy aims, such as economic growth and sustainable development, alongside scientific excellence (Lepori et al., 2023). This has resulted in a balancing act between the academic freedom most evident in Mode 1 and the public interest most evident in Mode 3. In between, Mode 2 is increasingly promoted in policy to enhance the societal relevance and impact of research, while also being a preferred approach in certain research traditions, for example, participatory research. There is a tradition among Nordic researchers of using such approaches - especially in research on working life and design - in line with the general Nordic model of cross-sectoral collaboration (Aagaard Nielsen & Svensson, 2006; Gunnarsson et al., 2016).

The effectuation of multiple, and partly competing, rationales in research policy has been studied in terms of principal-agent theory, where the "principals" of governments and research funding agencies set goals and priorities that are supposed to ensure that the "agents" of researchers and higher education institutions comply (de Jong et al., 2016; Lepori et al., 2023). A general conclusion is that even if there is a high incentive for such compliance in view of fierce competition for research grants, other incentives also affect agents' behavior in the form of scientific, professional, bureaucratic, legal and cultural standards. This leaves room for a wider spectrum of responses to research policy than acknowledged in principal-agent theory, ranging from compliance to coping, negotiation and dismissal (de Jong et al., 2016). It is suggested that the realization of intended policy outcomes depends on supportive practices by the research funder, on the one hand, and researchers' understanding and response to these practices, on the other (Norn et al., 2024).

It should also be noted that while excellence-focused funding is assumed to respect core academic values to greater extent than policy-focused funding, it tends to reinforce academic inequity and lack of diversity - "taking funding from less prestigious areas or research units to focus large sustained grants on research units already well performing" (Bégin-Caouette et al., 2017, p. 797). The rationale for policy-focused funding can thus be to compensate for adverse effects of established scientific norms with regard to current societal challenges, that is, "the failure of the research and innovation systems to engender the transformations necessary to generate and diffuse effective solutions to wicked societal problems" (Norn et al., 2024, p. 912). By extension, the scientific status and general familiarity of Mode 1 research could very well crowd out the generally less prestigious and familiar Mode 2 research, if the latter is not subject to earmarked funding. This is reflected in the limited literature on the practical design and implementation of public research funding to balance dual policy interests in excellent and open science (Norn et al., 2024).

To further understand how co-research is promoted and pursued in policy and research, we turn, lastly, to the four degrees of stakeholder participation distinguished in a European research review (Science Europe, 2017):

- *Cooperation* - with the lowest degree of collaboration, where stakeholders passively give researchers access to collection of data, for example, via their phones or estates.
- *Collaboration* - with a more active collaboration, where stakeholders contribute to collection of data on request from researchers, for example, by active observations of their surroundings.
- *Co-production* - with a more extensive collaboration, where stakeholders are involved also in the analysis of collected data on a more equal footing with researchers.
- *Co-design* - with the highest degree of collaboration, where stakeholders actively participate in the entire research process on equal footing with researchers.

The first two degrees - cooperation and collaboration – are, for example, characteristic of the research approach called citizen science, where citizens provide experimental data and facilities for researchers, most often in natural sciences (Lindberg, 2018; Science Europe, 2017). The other two degrees - co-production and co-design – are, in turn, characteristic of the research approach called participatory research, where stakeholders are actively involved in several parts of the research process, most often in social sciences (Aagaard Nielsen & Svensson, 2006; Gunnarsson et al., 2016). The very extent of stakeholder participation throughout the research process is considered crucial in co-research, with regard to four main phases: 1) Initiation, where stakeholders can contribute to identifying knowledge gaps and deciding research priorities, 2) Design, where stakeholders can contribute to formulating research questions and planning the research study, 3) Implementation, where stakeholders can contribute to collecting and analyzing data, as well as implementing and disseminating research results, and 4) Evaluation, where stakeholders can contribute to evaluating research results as a basis for future research priorities (Science Europe, 2017). A high degree and extent of stakeholder participation is often advocated in co-research policies and practices, despite the fact that each of the four degrees – as well as each of the three modes of academic knowledge production – can be appropriate for different research purposes.

Method and material

The method used in the study is a document analysis of call texts and applications in three of Forte's calls: Applied welfare research, 2022, Children's and young people's mental health, 2022, and Co-production for a sustainable working life, 2023. The material includes: 1) Call texts and appended instructions, 2) Metadata for approved and rejected applications to these calls with information about project title, summary, research subject and subject area, and 3) The entire application texts of approved projects in the calls.

Since the material is available in English and Swedish to various extents, the concepts used for co-research vary. For example, common English concepts for co-research in the call texts are collaboration and co-production, with Swedish equivalents being *samverkan* and *samskapande*. In this article, these are used in original English when available and translated from Swedish when not available in English. Several other concepts for co-research are also used in both calls and applications, in English and Swedish. These are used in original English as far as possible in the article and translated from Swedish when necessary.

The call texts and appended instructions were processed using Adobe Acrobat and Word, where content regarding concepts, motives, stakeholders and methods of co-research was identified by a full read-through. The metadata for all applications received

were processed using Excel and NVivo, where the sections on project title and summary were searched for the same contents as in the call texts. The application texts for approved applications were processed using Adobe Acrobat and NVivo, where the sections on summary, study design and collaboration/co-production with users were searched for the same contents as above. A random selection of five approved applications per call was processed, to ensure an equal number of processed texts from each call, and enable an in-depth examination of these. The selection included the first five applications on each list of approved applications.

The choice of elucidating content regarding concepts, motives, stakeholders and methods of co-research was based on its esteemed relevance for the study, and its congruence with the theoretical framework for multi-level relationships of co-research policy and practice. Finally, the processed data from each source was brought together in a comprehensive analysis of similarities and differences between demands for co-research in the calls and planned co-research in applications to these calls.

Co-research in calls and applications

This section presents the results regarding co-research in the calls and applications of Forte's three calls for funding: Applied welfare research, 2022, Children's and young people's mental health, 2022, and Co-production for a sustainable working life, 2023, referred to as Calls A, B and C in the following. A total of 286 applications were received in the calls, of which 42 were approved, and 244 were rejected.

Collaboration and co-production as dominant concepts

Collaboration and co-production are the main concepts used for co-research in Forte's three call texts. Both concepts are used in all calls, but with a preference for collaboration in Calls A and B, and co-production in Call C. In Call C, co-production is mentioned in the very title of the call – "Co-production for a sustainable working life" – and the call text includes a section titled "Co-production of research." In Calls A and B, there are corresponding sections titled "Collaboration/Co-production between research and practice."

In applications to the calls, collaboration is the most common concept for co-research in Calls A and B, appearing in 27% of summaries in Call A and 14% in Call B. Co-production is a more common concept in Call C, appearing in 26% of titles and 75% of summaries. Other related concepts also occur, for example, action research, interactive, practice-oriented, and participatory. Comparing approved and rejected applications, concepts linked to co-research occur to almost the same extent in titles for approved (12%) and rejected (14%) applications. In summaries, however, such terms occur more frequently in approved (67%) than in rejected (57%) applications. All approved applications studied in Call C, five in total, use the concept co-production, usually in the summary. In Call A, the term is used in two of five applications studied. In Call B, none of the applications use this concept, but collaboration is mentioned in two of five of them. All approved applications studied in Call A – five in total – use the concepts of client-oriented or practice-oriented research, in line with the call's general focus on promoting such research in the field of applied social services. Action research – and related concepts such as participatory research and interactive research – are also mentioned in some applications, in all calls.

The meaning of collaboration and co-production is explained to different extents in the call texts. In Call B, there are no explicit explanations, while it is explained in Call A that involvement of stakeholders only as study objects who contribute with data does not meet the requirements for collaboration and co-production. In Call C, it is explained that collaboration can mean consulting relevant stakeholders, appointing reference groups, and so on. Co-production, in turn, is explained in terms of stakeholders being actively involved in all or parts of the research process, from identifying research questions to jointly developing solutions and disseminating results. It is further clarified in the call that co-production is distinguished from other types of collaboration by its degree and extent of stakeholder involvement.

This variation is illustrated in approved applications, where a relatively high degree of collaboration is planned in an application to Call C, where stakeholders will be given a primary role in deciding the content of discussions as well as the practical knowledge and solutions developed, while a lower degree of collaboration is planned in an application in Call B, where stakeholders will be informed of the results via seminars and popular science reports.

Mandatory and optional co-research

Calls A and B state that the research must be planned and conducted in collaboration with relevant target groups. In Call B, however, this is only required when considered relevant for the research, as long as this is explicitly justified. Call C states that the research must be co-produced by researchers and users.

In all calls, engagement with end users is part of the assessment criteria for relevance. In Calls A and B, there are also guiding questions for assessments under the heading "Collaboration/co-production with (…) stakeholders in the research process." The questions are similar in both calls, and concern anchoring among stakeholders, how co-research affects the study design, co-research in different parts of the research process, and budgeted costs for co-research, among other things. In Call C, there are instead call-specific assessment criteria for co-research. These are less numerous than the questions in the other calls, but cover some of the same topics, and also the motive for co-research.

All calls include mandatory sections on co-research in their application forms. In Calls A and B, this section is titled "Engagement with end users." There, the applicants are asked to comment on the project's collaboration with stakeholders outside the scientific community. More specifically, they are asked to describe which actors will be engaged, in which parts of the research process and in what forms, as well as to motivate the relevance and describe expected effects of the collaboration. Call B adds that if collaboration or co-production is not relevant for the research project, this must be justified in the section on engagement with end users. In Call C, the corresponding section is titled "Co-production with end users." There, the applicants are asked to comment on how the co-production with relevant actors is planned and carried out. They are also asked to describe the same details as in the other calls. Calls A and C also include mandatory sections on previous experience of co-research in the research group.

Quality, relevance and utility as motives for co-research

All call texts state that the main motive for co-research is improved quality and relevance of research, without further explanation. In Call C, it is also stated that co-research

increases the likelihood of research results being utilized, and that it contributes to meeting challenges in working life. In the specific form of stakeholder collaboration called co-production, the intention is to "solve a complex problem where additional perspectives will increase the chances of finding a solution that will actually be implemented," according to the same call.

The approved applications describe motives for stakeholder collaboration to a varying extent – primarily in Calls A and C, and once in Call B. The most common motive is practical utility of research, stated in almost all applications studied in Calls A and C. The second most common motivation is quality of research, which occurs in three of five applications studied in Call A, two of five in Call C and one of five in Call B. In some applications, co-research is justified by relevance of research in relation to stakeholder needs, which is the case in two of five applications in Call A and one of five in Calls B and C. An example of described motives in approved applications is: "The creation of one project-wide reference group consisting of key stakeholders who can work together with the researchers to increase the quality in research by linking it closer to practice and everyday context" (Application in Call A).

Multi-sectoral stakeholder involvement

All calls mention examples of relevant stakeholders to involve in the research projects. Calls A and B mention, for example, user and interest organizations, professions, users and clients. Call A also states that the research projects must be conducted in collaboration with social services, in line with the call's general focus on promoting client-oriented and practice-oriented research in the field of applied social services. Call B asks, in turn, for asks for a description of involvement of children and young people in agreeing the approaches for collaboration or co-production, in line with the call's general focus on increasing knowledge about children's and young people's mental health. Call C mentions several examples of possible stakeholders, such as voluntary organizations, interest groups, labor-market parties or projects, professional practitioners, municipalities, regional health authorities and government agencies – also in line with the call's general focus on contributing knowledge that supports sustainable development of working life.

In approved applications, different types of stakeholders are planned to be involved in the research process. Most common are public sector actors and civil society organizations, who are involved in about two-thirds of the applications studied in all calls. Involvement of individual users and professionals is also common, which occurs in about half of the applications. The least common is involvement of actors from industry and business, which occurs in only one of the applications studied in Call C. Several applications involve multiple types of stakeholders. An example of described stakeholders in approved applications is:

"We will conduct guided 'stakeholder studios' in which we discuss openly with parents their needs and vision for the study. Additional studios will be conducted with family doctors, pediatricians, and child mental health providers" (Application in Call B).

Varying methods for co-research

All calls state that the more parts of the research process that stakeholders are involved in, the higher the level of collaboration. Descriptions are requested of how co-research will take place in the following parts: study design, data collection, interpretation, analysis, dissemination and implementation. Call C also mentions joint development of solutions to identified challenges and needs. In interpretation and analysis of data, co-research is

exemplified by discussing results and conclusions with stakeholders and making use of their reactions and opinions in continued research. Call C explains that co-production is distinguished from other types of collaboration by the very level of stakeholder involvement, by actively involving stakeholders in all or parts of the research process. In a similar vein, Call A underlines that only involving stakeholders as study objects who contribute data does not meet the requirements for collaboration or co-production. It is recognized in the calls that it may be difficult for applicants to specify methods for co-research on beforehand, since these are expected to be developed in dialogue with stakeholders. The applicants are nevertheless asked to describe the range of methods under consideration and the process for involving stakeholders in the project. In the associated instructions to the calls, it is stated that the applications must specify which parts of the research process and in what forms stakeholders will be involved.

In several of the approved applications, it is stated that stakeholder collaboration will take place in all parts of the research process. Specific parts are also mentioned, where the most common is study design – occurring in almost all approved applications in the calls. Other parts commonly mentioned are interpretation and analysis as well as collaboration in the development of new solutions – occurring in almost all applications in Calls A and C. Collaboration in implementation of research results is included in about half of the applications in all calls. Collaboration in dissemination is included in about a third of the applications. The parts with problem identification, data collection and results dissemination are mentioned in three of five applications studied in Call C. Overall, collaboration is planned in the most parts of the research process in approved applications in Call C, second most in Calls A and least in Call B. An example of described involvement in approved applications is: "(The stakeholders) will participate in all the phases of the research: from methods design (e.g., survey, focus groups, shadowing); to data collection; and discussion and diffusion of results through interactive workshops, seminar series and a conference" (Application in Call C).

Almost all approved applications studied describe their methods for stakeholder collaboration. A method commonly mentioned is involving stakeholders in steering groups, reference groups, or working groups. In some applications, relevant stakeholders are involved through other types of formations, such as a co-production team and a parent advisory committee. Another general method is to involve stakeholders through meetings, seminars, and workshops, such as co-production workshops and stakeholder studios. More specific methods for co-research are described in some applications, such as change agents and art-based embodied facilitation. In some cases, it is stated that planned methods for co-research are based on established approaches and traditions, such as action research, research circles, and participatory design. An example of described methods in approved applications is: "…we will invite end users (e.g., clinicians and patients) to our start up meeting and annual meetings, getting their inputs to help us optimize the research questions and designs, as well as sharing and discussing findings of the project" (Application in Call B).

Similarities and differences in calls and applications

A similarity between the three call texts is that they all encourage stakeholder collaboration in research, albeit in different extents and manners. All calls include assessment criteria for engagement with end users, and mandatory sections on co-research in their application forms. The concepts of collaboration and co-production are used in all call texts, and

also appear in titles and summaries of the applications received in all calls, as well as in the approved applications studied. We can here return to the different modes of academic knowledge production, presented in the theoretical framework (cf. Carayannis & Campbell, 2006; Nowotny et al., 2001). The demand for co-research in the calls seems most clearly related to Mode 2, where research is carried out in collaboration with stakeholders, based on research questions that are both theoretically and practically relevant. There is however no discussion in the calls that help situate co-research in a broader scientific landscape of different modes of academic knowledge production. This leaves applicants without guidance on how to esteem and prioritize co-research in relation to other possible approaches to address their research aims and areas.

Another similarity between the calls is that co-research is expected to contribute to improved quality and relevance of research. A difference is that practical utility is an additional motive in Call C, in terms of research results being utilized and contributing to meeting challenges in working life, with the intention to "solve a complex problem where additional perspectives will increase the chances of finding a solution that will actually be implemented." Utility is also the most common motive in approved applications in Calls A and C, followed by quality and relevance of research. In Call B, only one application studied states a motive for co-research, which is quality and relevance. Here, we can again return to the different modes of knowledge production, to situate these expected contributions in a broader scientific landscape. The quality of research has, for example, traditionally been associated with Mode 1, where little impact from the surrounding society is seen as a guarantee for objectivity of research. The relevance seems, in turn, related to Mode 2, in its focus on socially contextualized knowledge that is both theoretically and practically relevant. The practical utility seems related to Mode 3, where research helps understand and address societal challenges. The calls provide little guidance, however, on how to interpret and achieve these different contributions through co-research.

A difference between the calls concerns the strictness in demands for co-research in calls and the compliance of applications with these demands. Calls A and B both state that the research must be planned and conducted in collaboration with relevant target groups. In Call B, however, this only applies when considered relevant. In Call C, the research funded must be co-produced by researchers and users, which is further underlined by co-production being mentioned in the very title of this call. These differences are reflected in applications received, where collaboration or co-production is mentioned in 75% of summaries in Call C, and collaboration in 27% in Call A, but only in 14% in Call B. A similar pattern can be seen in the approval rate, where concepts related to co-research occur somewhat more frequently in approved than rejected applications. Overall, this means that demands for co-research in calls seem to correspond with planned use of this approach in applications. The data does not allow for affirming a causal relationship between co-research in calls and applications, however. Even if applicants formulate their applications in line with demands to increase their chances of funding, there might also be other factors at play, such as varying traditions of co-research among different researchers and disciplines.

Another difference concerns the degree of stakeholder collaboration requested in calls and planned in applications. As noted above, Calls A and B state that the research must be planned and carried out in collaboration with stakeholders (if relevant in Call B). Call C states, in turn, that stakeholders must be actively involved in all or parts of the research process. It is further emphasized in Call C that co-production is distinguished

from other types of collaboration by the very degree of stakeholder involvement. Approved applications include examples ranging from a low degree of collaboration where stakeholders are informed about the results via seminars and popular science compilations, to a high degree where stakeholders are given a prominent role in decisions about content and results. When comparing this with the different degrees of stakeholder participation outlined in the theoretical framework, some differences appear (cf. Science Europe, 2017). What is called collaboration in the calls seems to correspond with what is called co-production in the framework, with reference to stakeholder involvement not only in data collection but also in analysis. What is called co-production in the calls corresponds with what is called co-design in the framework, with reference to active stakeholder participation in the entire research process. The lowest degrees of stakeholder participation in the framework – cooperation where stakeholders passively give researchers access to data collection as well as collaboration where stakeholders contribute to data collection on request – are hardly considered in the calls, except for a clarification in Call A that involvement of stakeholders only as study objects does not meet the requirements for co-research.

All call texts state that the degree of collaboration increases the more parts of the research process that stakeholders are involved in. Parts mentioned in the calls are study design, data collection, interpretation and analysis, results dissemination and implementation – as well as joint development of solutions in Call C. In approved applications, collaboration is planned in the most parts of the research process in Call C, the second most in Call A and the least in Call B. Collaboration is most common in study design, followed by interpretation and analysis, as well as development of solutions. When comparing this with stakeholder participation in different parts of the research process outlined in the theoretical framework, some differences appear (cf. Lindberg, 2018; Science Europe, 2017). The framework extends the process at both ends of the range mentioned in the calls, by including initiation and evaluation as well. In between, the framework only mentions two parts, design and implementation, while the calls mention several, including study design, data collection, interpretation, analysis, dissemination, implementation and solutions development. The calls are less specific regarding practical methods for co-research in different parts of the research process, which is explained by these being developed later to enable stakeholder dialogue. This seems contradictory, however, since this ought to apply to all project activities where collaboration is planned. Not least since almost all of the approved applications studied describe their planned methods for co-research.

Discussion

The study reveals that demands for co-research in Forte's calls correspond with planned use of this approach in received and approved applications. In the calls where co-research was mandatory, in combination with a demand for active stakeholder involvement in several parts of the research process, this is reflected to a greater extent in applications compared with calls where it was voluntary or less far-reaching. This seems to confirm the assumption in principal-agent theory that the goals and priorities set by governments and research funding agencies ensure that researchers comply or, at least, cope (cf. de Jong et al., 2016; Lepori et al., 2023). The compliance does not necessarily prove an unambiguous causality, however, since there might be other factors at play than applicants' compliance with calls, such as their possible interest in and tradition of conducting

co-research (cf. Aagaard Nielsen & Svensson, 2006; Gunnarsson et al., 2016). There are some indications of a wider spectrum of responses among applicants beyond compliance, where previous research distinguishes negotiation and dismissal as well (de Jong et al., 2016). One could, for example, argue that negotiation is perceivable in practical utility being the most common motive for co-research in approved applications in two of the calls, and in the descriptions of planned methods for co-research in almost all of the approved applications studied, while being less prominent in the call texts.

Another finding is that there are few discussions in the calls that help situate co-research in a broader scientific landscape of different modes of academic knowledge production (cf. Carayannis & Campbell, 2006; Nowotny et al., 2001). There is also a lack of discussion on the possible appropriateness of the lower degrees of stakeholder participation, that is, cooperation and collaboration, than the ones targeted in Forte's promotion of the higher degrees, that is, co-production and co-design (cf. Science Europe, 2017). An additional gap is the lack of discussion on how to interpret and achieve the improvements in scientific quality, as well as societal relevance and impact, that are expected benefits of co-research. Yet another gap is the lack of discussion on different traditions and methods for co-research, despite the presence of such descriptions in almost all approved applications studied (cf. Aagaard Nielsen & Svensson, 2006; Lindberg et al., 2019). Taken together, this leaves applicants without guidance on how to esteem, prioritize, design and conduct co-research, with regard to both other modes of academic knowledge production in general, and different modes of co-research in particular.

By extension, the lack of guidance can make the balancing act between the co-existing rationales of academic freedom and public interest in public research funding (cf. Lepori et al., 2023) more difficult, for both researchers and research funders. This is because it may counteract their potential upskilling that may otherwise be the result of earmarked calls for co-research. Such upskilling especially concerns promoting and conducting co-research in ways that achieve the expected improvements in the quality, relevance, and impact of research. Such upskilling may be necessary for Mode 3 research being able to compensate for the adverse effects of the generally high status of Mode 1 and less prestigious Mode 2 (cf. Bégin-Caouette et al., 2017). This is in line with previous findings that the realization of intended policy outcomes depends both on supportive practices by the research funder, and on researchers' understanding and response to these practices (cf. Norn et al., 2024). This can be further understood in light of the two-directional relationship between innovative institutions, processes and solutions, distinguished in previous research on public sector innovation (cf. Ansell & Torfing, 2022; Sørensen & Torfing, 2022).

Implications

The study helps advance understanding of the relationship between policies and practices in public promotion of co-research, by highlighting the relationship between the different levels of public sector innovation (Sørensen & Torfing, 2022) in the ongoing academic reorientation towards a wider and deeper societal engagement (Brundenius et al., 2017). This primarily concerns the relationship between the policy level, corresponded by Forte's encouragement of co-research in their calls for funding, and the process level, corresponded by the planned co-research in applications to these calls. This relationship is shown to be strong in the sense of being dominated by compliance, or at least coping, among applicants, although with some room for negotiation (cf. de Jong et al., 2016).

At the same time, the relationship is vague in the sense of applicants being largely left without guidance on how to esteem, prioritize, design and conduct co-research, with regard to other modes of academic knowledge production in general and different modes of co-research in particular (cf. Carayannis & Campbell, 2006; Nowotny et al., 2001; Science Europe, 2017). This ambiguous relationship between co-research policy and practice might have implications for the third level of public sector innovation, that is, the institutional designs that facilitate the scaffolding of policies and processes (cf. Sørensen & Torfing, 2022). This is because it further complicates the overarching balancing act between the co-existing rationales of academic freedom and public interest in public research funding, for both researchers and research funders (cf. Lepori et al., 2023).

Our findings suggest that at a time when academic freedom is subject to public pressure from both governments and citizens, it could be safeguarded by intermediaries in the form of governmental research council – as well as communities in the form of research groups and higher education institutions – with skills in navigating different modes of knowledge production and different modes of co-research (cf. Norn et al., 2024). Here, we can return to research on public sector innovation, concluding that "the advancement of public innovation not only requires creativity and courage, but also key competences in stakeholder analysis, facilitation and institutional design" (Sørensen & Torfing, 2022, p. 36). For research funders, the ideal trade-off between respecting core academic values and compensating for adverse effects towards co-research (cf. Bégin-Caouette et al., 2017) is of course difficult to pinpoint in detail. For researchers and their institutions, the ideal mix of responses across the spectrum of compliance, coping, negotiation and dismissal (cf. de Jong et al., 2016) is also tricky to pinpoint, in view of their legal obligation in Sweden to both collaborate with the surrounding community and to promote and protect their academic freedom (cf. The Swedish Higher Education Act 1992:1434). Some guidance can nevertheless be found in previous suggestions that institutional support of collaborative innovation processes "must be socially embedded through a process of reflexive adaptation so that they fit the social, political, and economic conditions on the ground" (Sørensen & Torfing, 2022, p. 45), which is why Forte conducts the learning process that forms the basis for our study.

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